Mat 2355 Test.

October 20 2004 length 1: 20

Please **print** your name and student number on the booklet. Could you please also indicate the **order** in which you answer the questions on the booklet cover. You must justify your answers.

1. (6) Let ΔABC be a triangle in the Euclidean plane $\mathbf{E}^2$. Show that $\angle A + \angle B + \angle C = 180°$, where $\angle A$,$\angle B$ and $\angle C$ are the angles at the corresponding vertex.

2. (6) Fix a point O in the Euclidean plane. Let f be the symmetry of $\mathbf{E}^2$ with respect to O, that is, $f(P) = Q$ if and only if $\overrightarrow{OQ} = -\overrightarrow{OP}$.

a) Show that f is an isometry. You may assume that reflections (with respect to lines), rotations and translations are isometries.

b) Write f as a product (composition) of reflections.

3. (6) Let $ABCD$ be a parallelogram and $M = \overline{AC} \cap \overline{BD}$ the intersection of the diagonals. Show that M is the midpoint of $\overline{AC}$.

4. (7) Let A, B be two distinct points on a circle C centered at O. Assume that the tangents to C at A and B intersect at the point P.

a) Show that $\overline{AB}$ est perpendicular to $\overline{OP}$.

b) Show that $\angle PAB = \frac{1}{2}\angle AOB$.